

Event-Guided Dynamic Reconstruction under Extreme Motion Blur

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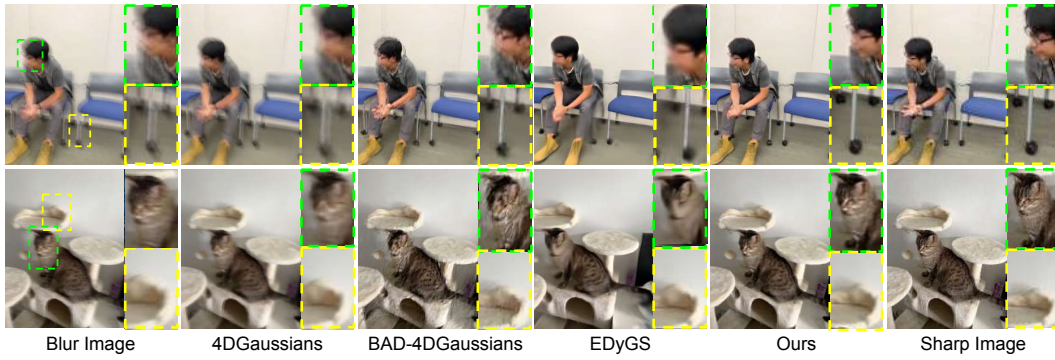


Figure 1: Our proposed framework effectively reconstructs sharp novel views using event data from monocular videos affected by extreme motion blur.

Abstract

Dynamic scene reconstruction relies on accurate feature correspondences across frames. Under motion blur, these correspondences become unreliable, causing SfM pipelines to fail. To address this challenge, we propose a framework for event-guided dynamic reconstruction under extreme motion blur that leverages the high temporal resolution of event cameras to guide camera trajectory estimation and scene deblurring. By fusing asynchronous event data with blurry images, our method recovers sharp dynamic scenes from blurred images. To model smooth, continuous motion, we represent the camera trajectory as continuous B-splines, jointly optimized with dynamic 3D Gaussians. Experiments show that our method improves PSNR by 0.52 dB for deblurring and 1.02 dB for novel view synthesis under the most challenging blur level in our evaluation. The proposed continuous trajectory formulation further reduces pose error by 35% compared to discrete alternatives. By integrating event-based sensing with continuous trajectory modeling, our framework enables robust dynamic scene reconstruction under extreme motion blur. ¹

1 Introduction

Reconstructing accurate scenes from images is central to applications in augmented reality, robotics, traffic monitoring [3, 43] and digital content creation. However, captures for real-world reconstruction

¹ Video results: <https://figshare.com/s/35892b02f9641ea8e46c>

are often monocular video captures suffering from severe motion blur from moving objects and sporadic handheld camera motion. This blur disrupts feature correspondences, leading to geometric inconsistencies and frequent pose estimation failures in standard reconstruction pipelines.

Recent works [30, 36, 42, 56] on blur-aware reconstruction jointly optimize per-frame camera trajectories by learning adjustments to the initially estimated poses. However, these works rely on initial pose estimates for subsequent optimization, which are often unavailable or unreliable under severe motion blur. Further, their fragmented per-frame modeling does not explicitly account for the temporal continuity of camera motion, resulting in discontinuous pose transitions between consecutive frames (see Figure 2) and degraded reconstruction quality. Given these limitations of conventional cameras, event cameras have gained significant attention for their potential to overcome the challenges of frame-based imaging. Event cameras asynchronously capture per-pixel brightness changes with high dynamic range and microsecond latency, providing temporally dense motion information even when frame-based features vanish. This makes them inherently robust to motion blur and well-suited for dynamic scenes where conventional sensors fail to record sharp features [44, 45].

In this work, we exploit event data to guide camera pose initialization in extreme motion-blur scenarios, enabling robust reconstruction even when traditional Structure from Motion (SfM) pipelines like COLMAP [40] fail. By leveraging event data, we recover sharp intensity information from blurry images, restoring fine image features that allow SfM to estimate camera poses at a higher frequency than the capture rate of the original blurred frames. To further enhance temporal consistency of camera optimization, we parameterize the camera trajectory as continuous cubic $SE(3)$ B-splines. This continuous representation enforces smooth motion and reduces the number of parameters, compared to per-frame modeling. We initialize the spline using event-guided SfM trajectory and jointly refine it through photometric optimization over both blurred frames and reconstructed sharp intensities. In this continuous formulation, the local support property of B-splines ensures that each control point influences a temporal neighborhood. As a result, refining a camera pose affects nearby poses, naturally enforcing temporal coherence and producing a globally consistent camera trajectory.

Our method significantly improves reconstruction quality, as shown in Figure 1. Event-guided supervision alone yields up to 2.33 dB in PSNR over the corresponding frame-only baseline, and our continuous pose module provides a further 1.02 dB improvement in novel view synthesis. In addition, our framework demonstrates strong robustness under increasing levels of motion blur. To summarize, the main contributions of our work are outlined below:

1. We introduce a novel, continuous camera trajectory formulation based on cubic $SE(3)$ B-splines, which naturally enforces temporal smoothness and globally optimizes the entire trajectory.
2. We present a framework that integrates asynchronous event data with a dynamic scene reconstruction pipeline that remains operable and maintains reconstruction quality even in extreme motion blur scenarios.
3. Our approach consistently outperforms both frame-based and event-guided baselines in scene reconstruction and optimized pose estimation. Specifically, we reduce pose estimation error by 35% and yield PSNR improvements of 0.52 dB in deblurring and 1.02 dB in novel view synthesis.

2 Related Works

Scene Reconstruction under Motion Blur. Recent approaches have addressed the challenge of static scene reconstruction from motion-blurred images by extending neural radiance fields and Gaussian Splatting formulations. Deblur-NeRF [31] models the exposure process as a temporal integration of radiance to jointly estimate spatially varying blur kernels. Building on this concept, PDRF [35] and BAGS [36] learn blur kernels for blur-agnostic representation, while SMURF [21] introduces a neural ordinary differential equation (ODE) formulation to represent continuous camera motion.

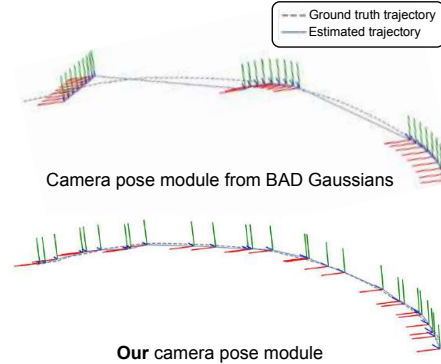


Figure 2: Visualization of the first 30 estimated camera poses on the *sriracha-tree* scene.

Other works [20, 46, 47, 56] jointly refine camera poses and scene geometry by parameterizing per-frame camera motion using linear, cubic, or Bézier trajectories. To handle dynamic environments, DyBluRF [42] introduces flow and depth supervision, whereas BARD-GS [30] incorporates guidance from 3D trajectories. Typically, these methods update each refined pose via $\mathbf{P} \leftarrow \mathbf{P}_{\text{est}} \times \Delta\mathbf{P}$, where the per-frame pose adjustment $\Delta\mathbf{P}$ is parameterized using local cubic splines for individual frames. However, such independent, frame-wise modeling leads to fragmented motion representations that break global temporal consistency, resulting in discontinuous camera trajectories when jointly optimized alongside scene parameters. In contrast, we model camera motion as a single, globally continuous $\text{SE}(3)$ B-spline trajectory, thereby enforcing temporal smoothness and reducing parameter redundancy. Furthermore, while existing methods rely solely on degraded RGB frames, which causes both pose tracking and deformation learning to suffer from input degradation, our approach inherently leverages asynchronous event data. These high frequency motion cues directly complement the frame based inputs, effectively regularizing the optimization to recover sharp scene structures and stabilize overall pose estimation.

Scene Reconstruction using Events. Complementing RGB-based techniques, event-driven methods have been explored for high speed and motion robust scene reconstruction. Several frameworks adapt neural radiance fields [14, 18, 32, 39] and Gaussian Splatting [8, 49, 51] to event streams, with specific variants incorporating sensor non idealities [27] and enabling static reconstruction without predefined poses [11]. However, these frameworks primarily focus on clean event data and do not explicitly address the challenges associated with scene reconstruction with blurred images.

To enhance robustness to motion blur, event-guided methods combine event streams with conventional frames for static scene reconstruction using event-based supervision [2, 6, 37]. For instance, certain approaches [23, 33] recover 3D structure and camera trajectories from a single blurry image, while Deblur-e-NeRF [28] models physically accurate pixel bandwidths to alleviate blur artifacts. To optimize geometry and camera trajectories simultaneously, EvaGaussians [54] utilizes recovered intensities derived from the Event-based Double Integral (EDI) formulation [34], whereas DeblurSplat [22] enables static reconstruction without initial pose information. Although recent methods extend fragmented per-frame pose optimization to both static [7, 25] and dynamic scene reconstruction [29], they inherit the fundamental limitation of per-frame fragmented pose updates, which lack strict temporal continuity. Our framework bridges this gap by unifying event-guided dynamic deblurring with a continuous B-spline trajectory formulation, ensuring stable and temporally consistent dynamic scene reconstruction under extreme motion blur.

Method	Dyn.	Deblur	Event	Traj. Rep.
Deblur-NeRF [31]	✗	✓	✗	-
4DGaussians [48]	✓	✗	✗	-
BAD-Gaussians [56]	✗	✓	✗	piecewise
BARD-GS [30]	✓	✓	✗	piecewise
EventNeRF [39]	✗	✗	✓	-
EvaGaussians [54]	✗	✓	✓	piecewise
EDyGS [29]	✓	✓	✓	discrete
Ours	✓	✓	✓	Continuous

Table 1: Our approach uniquely unifies event guidance with globally continuous trajectory optimization.

3 Background: Motion Blur Formulation and Event-based Double Integral

A conventional camera forms an image by accumulating incident photons over an exposure duration. When the camera or the scene undergoes motion during this interval, the resulting image suffers from motion blur. This image formation process can be mathematically modeled as the temporal integration of a sequence of sharp latent images over the exposure duration. Specifically, for each pixel location $\mathbf{u}$, the observed blurred color value $\tilde{C}_{\mathbf{u}}$ is given by:

$$\tilde{C}_{\mathbf{u}} = \frac{1}{\Delta t} \int_{t_0}^t C_{\mathbf{u}}(t') dt' \approx \frac{1}{N} \sum_{i=0}^{N-1} C_{\mathbf{u}}(i), \quad (1)$$

where $C_{\mathbf{u}}(t')$ represents the color at pixel $\mathbf{u}$ in the sharp latent image at time t' , and $\Delta t = t - t_0$ is the total exposure time. In practice, this continuous integration is typically approximated by a discrete sum over N uniformly sampled latent images.

In contrast, an event camera captures an event $\mathbf{e} = \langle \mathbf{u}, p, t \rangle$ asynchronously, with polarity $p = \pm 1$ at time t and pixel $\mathbf{u}$ when the change in log intensity exceeds the threshold ρ . The pixel intensity $I_{\mathbf{u}}(t)$ at time t can be estimated using the sum of events $E_{\mathbf{u}}(t - t_0)$ from time t_0 to t ,

$$\log I_{\mathbf{u}}(t) - \log I_{\mathbf{u}}(t_0) = \rho E_{\mathbf{u}}(t - t_0) \implies I_{\mathbf{u}}(t) = I_{\mathbf{u}}(t_0) \exp(\rho E_{\mathbf{u}}(t - t_0)). \quad (2)$$

This formulation expresses pixel intensity at time t as a function of intensity at time t_0 and the accumulated events. By applying the blur model (Equation 1) to pixel intensities, the observed blurred intensity $\tilde{I}_{\mathbf{u}}$ and events in the exposure interval $[t_0, t]$ can be used to formulate the sharp latent intensity at t_0 ,

$$\begin{aligned} \tilde{I}_{\mathbf{u}} &= \frac{I(t_0)}{\Delta t} \int_{t'=t_0}^t \exp(\rho E(t' - t_0)) dt' \\ \implies \log I(t_0) &= \log \tilde{I}_{\mathbf{u}} - \log \left(\frac{1}{\Delta t} \int_{t'=t_0}^t \exp(\rho E(t' - t_0)) dt' \right). \end{aligned} \quad (3)$$

Thus, Equation 3 establishes a linear relation between latent and blurred pixel intensities in the logarithmic space. By combining Equation 2 and 3, we can estimate the sharp latent pixel intensities using a blurred image and event data corresponding to its entire exposure interval. Building upon these formulations, the next section presents our blur-aware dynamic Gaussian Splatting framework for dynamic scene reconstruction from blurred RGB-event data.

4 Methodology

We decompose our method into three components: *(i)* event-guided initialization for robust pose estimation under extreme blur, *(ii)* a continuous-time camera trajectory representation using cubic B-splines in $\text{SE}(3)$ jointly optimized with scene parameters, and *(iii)* a multi-stage deblurring framework that uses event-recovered intensities and blurry images as supervision for dynamic scene reconstruction.

4.1 Event-aided Pose Estimation

We address motion blur by integrating high-temporal-resolution event data into the SfM initialization process, bridging the gap between asynchronous events and frame-based reconstruction.

We use EDI (Eq. 3) to reconstruct temporally dense, edge-preserving intensity images from asynchronous event streams and corresponding blurred frames. These reconstructions replace raw RGB inputs during SfM initialization, enabling feature detection and triangulation in cases where conventional COLMAP fails. Unlike prior event-assisted SfM approaches [23, 54], which refine sparse poses at the frame rate of blurry images, our module estimates temporally continuous camera trajectories from EDI-recovered frames, providing a high-frequency initialization.

4.2 Camera Pose Module

To achieve temporally consistent motion modeling, our camera pose module represents the entire camera trajectory as a single continuous cubic B-spline curve in $\text{SE}(3)$ and jointly optimizes it with the 3D Gaussians. This formulation learns a global continuous curve parameterized by shared control points $\mathbf{Q}_i$, enforcing cross-frame coherence and temporal consistency across the sequence. The trajectory is defined by a B-spline with n segments and degree $d = k - 1$, parameterized by $n + k - 1$ control points $\mathbf{Q}_i$. The number of segments n is selected empirically based on the observed motion complexity. The pose at time t , $\mathbf{P}(t)$, is then obtained by sampling the continuous curve as follows:

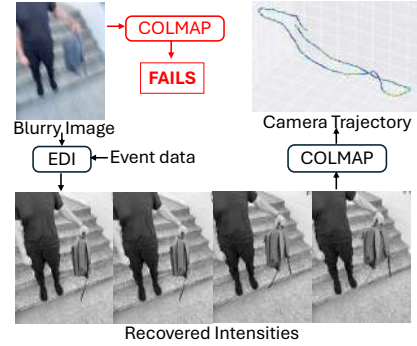


Figure 3: **Event-aided pose estimation pipeline.** Our event-guided initialization restores structural edges lost in blur, enabling reliable pose estimation where conventional SfM collapses.

$$\mathbf{P}(t) = \prod_{i=0}^{n+k-2} \exp(N_{i,k}(t) \log \mathbf{Q}_i). \quad (4)$$

Here, $N_{i,k}(t)$ are B-spline basis functions defined recursively using the Cox-de Boor formula [5]. These basis functions determine the influence of each control point $\mathbf{Q}_i$ on the camera pose $\mathbf{P}(t)$ and are defined over a non-decreasing knot vector $\{t_0, t_1, \dots, t_{n+k-2}\}$, which partitions the time domain and controls the support and continuity of the B-spline. Each basis function $N_{i,k}(t)$ has support over the interval $[t_i, t_{i+k})$, implying that the pose $\mathbf{P}(t)$ is influenced by $k+1$ control points characterizing the local support property of B-splines.

A B-spline curve of degree $d = k - 1$ guarantees C^{d-1} continuity, ensuring that the trajectory is differentiable up to order $d - 1$. Even with $d = 1$, the resulting trajectory has C^0 continuity and remains piecewise continuous. The camera poses corresponding to adjacent time intervals share k control points, promoting smooth trajectory modeling across the time intervals. A latent sharp image can be rendered for each camera pose $\mathbf{P}(t)$ using the dynamic Gaussian framework of 4DGaussians [48] (Appendix A.1), establishing a one-to-one correspondence between poses and rendered images. $\mathbf{P}(t)$ and control points $\mathbf{Q}_i$ are differentiable with respect to these rendered images, allowing end-to-end optimization during training. We start by fitting a B-spline to the estimated camera poses. This trajectory is subsequently refined through joint optimization with the canonical Gaussians $\mathbb{G}_c$, which ensures pose adjustment during the training process.

4.3 4D Deblurring Framework

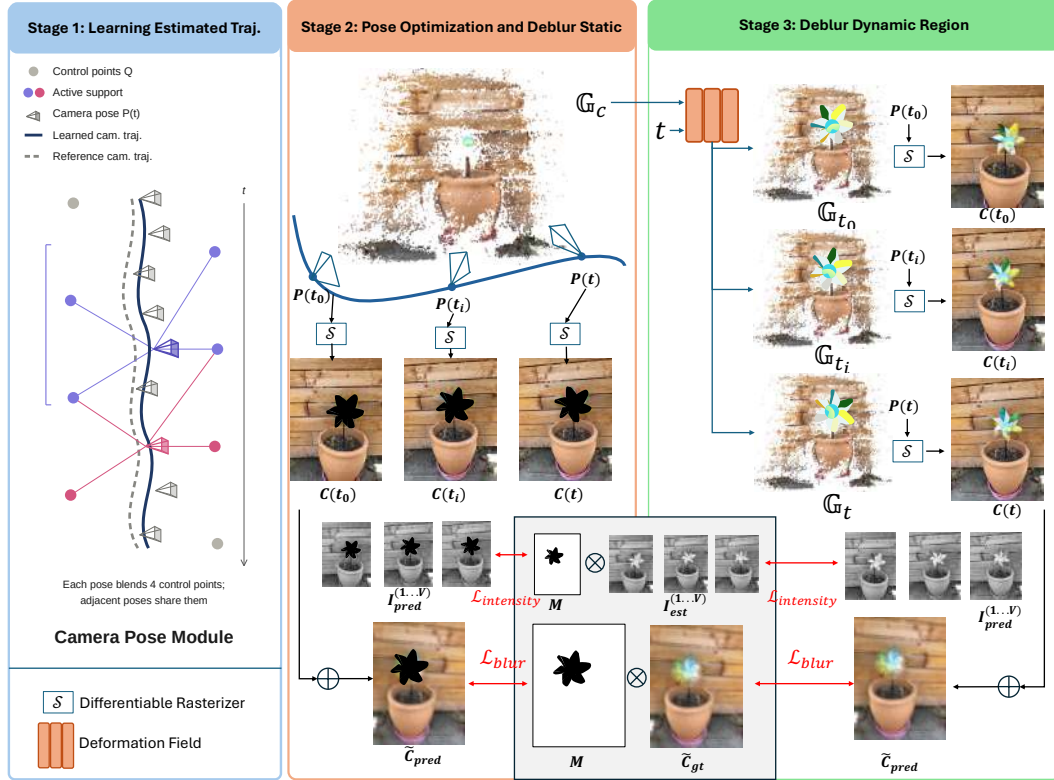


Figure 4: **Overview of the proposed framework.** Stage 1: optimizes a continuous camera trajectory using B-spline control points in $\text{SE}(3)$. Stage 2: refines the trajectory using event-based supervision and photometric losses restricted to static regions via dynamic masking. Stage 3: models object motion with a learned deformation field for dynamic scene reconstruction.

We introduce a three-stage training framework. Stage 1 learns an initial trajectory by aligning B-spline parameterized poses $\mathbf{P}(t)$ with EDI-based pose estimates $\mathbf{P}_{\text{est}}(t)$ by minimizing the geodesic distance between the two over N camera poses:

$$\mathcal{L}_{\text{pose}} = \frac{1}{N} \sum_{i=1}^N \|\text{Log}(\mathbf{P}(t_i) \cdot \mathbf{P}_{\text{est}}(t_i)^{-1})\|_2, \quad (5)$$

where $\mathbf{P} \cdot \mathbf{P}_{\text{est}}^{-1}$ is the residual transformation, and $\text{Log}(\cdot)$ maps to the Lie algebra $\mathfrak{se}(3) \cong \mathbb{R}^6$, whose ℓ_2 norm gives the scalar geodesic distance.

In Stage 2, the B-spline camera trajectory is jointly refined with the static part of the scene with photometric supervision from both blurry RGB frames and event-recovered sharp intensities. Given the exposure interval Δt , we sample V intermediate poses $\mathbf{P}_v(t)$ from the spline and render corresponding sharp images using static canonical Gaussians. To isolate the static scene content, we apply a dynamic region mask M , ensuring that gradients are computed only for static Gaussians. We obtain the synthesized blurry image by averaging the V rendered images, $\tilde{\mathbf{C}}_{\text{render}} = \frac{1}{V} \sum_{v=1}^V \mathbf{C}_{\text{render}}^{(v)}$, and compute a photometric loss between $\tilde{\mathbf{C}}_{\text{render}}$ and the observed blurry image $\tilde{\mathbf{C}}_{\text{gt}}$. Additionally, we convert the rendered images to grayscale intensities $\mathbf{I}_{\text{render}}^{(1 \dots V)}$ and compute photometric loss against recovered intensities $\mathbf{I}_{\text{est}}^{(1 \dots V)}$.

The masking M constrains gradients to static Gaussians, ensuring that pose optimization is unaffected by object deformations. This selective supervision prevents spurious gradients from dynamic regions and stabilizes camera trajectory refinement. The total loss is given by,

$$\mathcal{L}_{\text{total}} = \lambda_{\text{blur}} \cdot \left\| M \odot (\tilde{\mathbf{C}}_{\text{gt}} - \tilde{\mathbf{C}}_{\text{render}}) \right\|_1 + \lambda_{\text{intensity}} \cdot \sum_{v=1}^V \left\| M \odot (\mathbf{I}_{\text{render}}^{(v)} - \mathbf{I}_{\text{est}}^{(v)}) \right\|_1. \quad (6)$$

In Stage 3, we fix the optimized camera trajectory and learn a deformation field $\mathcal{F}$ that models non-rigid object motion. Rendered views are averaged to simulate motion blur, and a photometric loss is computed over the entire image without masking, allowing optimization over both static and dynamic regions. The deformation field uses a multi-resolution hexplane representation and a lightweight MLP similar to 4DGaussians [48], enabling temporally coherent reconstruction across dynamic regions while preserving the sharpness achieved in static areas.

Collectively, these stages form a cohesive optimization pipeline that fuses event-driven supervision with continuous trajectory modeling and Gaussian-based reconstruction, enabling accurate recovery of dynamic scenes under motion blur.

5 Experiments

Dataset. We construct a synthetic dataset comprising paired blurry images and event data for evaluation in a controlled setting. Starting from video sequences recorded at 30 *fps*, we upsample them to 240 *fps* using the pretrained video interpolation model RIFE [13]. The high-frame-rate sequences are then used to generate event data using ESIM [38]. Following the procedure of [37], motion blur is synthesized by temporally averaging consecutive high-frame-rate frames. To model varying degrees of motion blur, we create three levels of blur: medium (MB), severe (SB), and extreme (XB), corresponding to averages over 24, 40, and 72 consecutive frames, respectively.

The dataset includes eight scenes from the DyCheck [10] dataset. Using only the blurry images, COLMAP fails to recover camera poses for all scenes at the XB (extreme blur) level and SB (severe blur) level. In contrast, our event-guided approach successfully recovers accurate poses across all blur levels. For evaluation, we use the camera poses from our pipeline for all methods. For evaluating novel view synthesis, every 10th frame from the original sequence is held out for novel view evaluation. Further dataset details are provided in the Appendix A.2.

Baselines. We evaluate our method against a comprehensive set of state-of-the-art baselines for dynamic scene reconstruction under motion blur, structured across three main categories. First, for standard dynamic frameworks, we compare against 4DGaussians [48] (4DGau.) and ED3DGS [1]. Second, we evaluate scene deblurring approaches, including 4DGau.+MPRNet [55] to assess pre-deblurring with a pretrained network, DyBluRF [42], a NeRF explicitly designed to handle dynamic blur, and BAD-4DGau., an extended version of BAD-Gaussians [56] adapted for 4D dynamic scenes.

Methods	Deblurring			Novel View Synthesis		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	SI-PSNR $\uparrow$	SI-SSIM $\uparrow$	SI-LPIPS $\downarrow$
ED3DGS [1]	19.59	0.5244	0.5365	18.13	0.5071	0.5521
4DGau. [48]	19.85	0.5321	0.5398	18.68	0.5129	0.5536
4DGau. + MPRNet [55]	19.66	0.5258	0.5344	18.69	0.5120	0.5442
DyBluRF [42]	17.42	0.2717	0.6939	16.54	0.2597	0.7022
BAD-4DGau.	18.65	0.4592	0.4866	17.88	0.4498	0.4961
4DGau. (w/ EV)	20.35	0.5467	0.5199	19.07	0.5241	0.5322
BAD-4DGau. (w/ EV)	20.98	0.5488	0.4358	19.40	0.5170	0.4556
EDyGS [29]	21.31	0.5881	0.5058	19.07	0.5360	0.5227
Ours	21.83	0.5755	0.4091	20.42	0.5503	0.4244

Table 2: Quantitative comparison under extreme blur (XB). SI-* are shift-invariant metrics. All baselines use camera poses from our event-guided estimation. **Best**, **second**, and **third** results.

Finally, to thoroughly examine the influence of various supervisory components, we introduce a third set of baselines with event-based supervision. This includes 4DGau. (w/ EV), BAD-4DGau. (w/ EV), EDyGS [29], and **Ours**. 4DGau. (w/ EV) incorporates event-based intensity supervision and blur loss without the pose refinement. EDyGS is designed for color event streams, which are unavailable in our setting. We add blur loss on the captured blurry RGB frames to recover the missing color supervision. BAD-4DGau. (w/ EV) serves as a controlled baseline that differs from **Ours** only in the pose refinement module, enabling a direct evaluation of the effectiveness of our proposed pose refinement approach.

Evaluation Metrics. For deblurring evaluation, we report standard image quality metrics, Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS). For novel view synthesis, we employ shift-invariant (SI) metrics introduced in [30], which allows a 10% translation along both horizontal and vertical axes, reporting the best score across shifts. This adjustment compensates for small spatial misalignment between rendered and ground-truth sharp images, preventing unfair penalization during pixel-wise comparison.

Implementation Details. We first train the proposed camera pose module to obtain a single camera trajectory supervised by the estimated camera poses. In this stage, the optimal number of B-spline segments is empirically determined. Initialized at 200 segments, the count is iteratively incremented by 40 until the mean pose error falls below 5×10^{-3} , balancing model expressivity for specific camera trajectory. Because this stage optimizes only a small set of spline control points, it adds negligible overhead relative to the subsequent Gaussian optimization. The training then proceeds in two stages: Stage 2 for 10,000 iterations and Stage 3 for 14,000 iterations. To ensure a fair comparison, all baseline methods are trained following the same procedure and training schedule. DyBluRF is trained for 30,000 iterations. All experiments are conducted on an NVIDIA A100 GPU with 80 GB of memory.

5.1 Deblurring and Novel View Synthesis

We evaluate all methods under the extreme blur (XB) setting, using camera poses obtained from our pipeline for consistency across comparisons as COLMAP fails to estimate camera poses on images with this magnitude of blur. Reported results represent the average performance across all evaluated scenes. Video comparisons across all scenes are available at the link in footnote ¹.

As shown in Table 2, methods operating without event-guided supervision under the extreme motion (XB) setting yield relatively uniform baseline performance, with pre-deblurring (4DGau. + MPRNet [55]) performing similarly to standard 4DGau. [48]. Optimization-based deblurring frameworks such as BAD-4DGau. and DyBluRF [42] face under-constrained conditions, where joint pose and appearance estimation can become counterproductive without robust supervision. Incorporating event-guided supervision significantly boosts reconstruction quality, improving deblurring PSNR by 0.50 dB for 4DGau. and 2.33 dB for BAD-4DGau. EDyGS [29] attains the highest deblurring SSIM (0.5881) but trails in PSNR and LPIPS and across all shift-invariant metrics, suggesting that its reconstructions preserve structure without recovering fine photometric detail. Our full model (**Ours**) achieves the best overall performance. Against BAD-4DGau. (w/ EV), which differs from our method only in the pose refinement module, we improve by 0.85 dB in deblurring PSNR, 1.02 dB in

SI-PSNR, and ≈ 0.03 in both SSIM metrics. This validates that integrating event-guided intensity supervision with our proposed pose module enhances reconstruction fidelity and temporal consistency under severe motion.

Methods	Deblurring			Novel View Synthesis			APE (cm)↓
	PSNR↑	SSIM↑	LPIPS↓	SI-PSNR↑	SI-SSIM↑	SI-LPIPS↓	
MLP-4DGau. (w/ EV)	19.18	0.4925	0.4980	18.84	0.4925	0.5011	1.8937
BAD-4DGau. (w/ EV)	20.98	0.5488	0.4358	19.40	0.5170	0.4556	1.9845
EDyGS [29]	21.31	0.5881	0.5058	19.07	0.5360	0.5227	1.8270
<i>Ours (linear)</i>	21.36	0.5683	0.4183	19.11	0.5425	0.4500	1.5238
<i>Ours (cubic)</i>	21.83	0.5755	0.4091	20.42	0.5503	0.4244	1.1867

Table 3: Comparison of camera pose formulations under extreme blur (XB): an MLP-based parameterization [50], per-frame optimization [56], per-pose optimization [29], and our continuous B-spline trajectory with linear and cubic variants. **Best** and **second** results.

5.2 Comparison of Camera Pose Formulations

To further assess the effectiveness of our proposed pose optimization module, we compare it against alternative pose paradigms: an MLP-based approach adapted from Deblur4DGS [50] (*i.e.*, MLP-4DGau. (w/ EV)), BAD-4DGau. (w/ EV), and EDyGS [29], which optimizes each EDI-estimated pose independently. We evaluate our formulation using both linear and cubic B-spline trajectories under the extreme blur (XB) setting to analyze the effect of spline order.

As shown in Table 3, our continuous B-spline formulations outperform the discrete and per-pose baselines on nearly all metrics, with the cubic variant achieving the best result in every category except SSIM, where EDyGS is marginally higher (0.5881 vs. 0.5755). The trajectory metrics make the source of these gains explicit. *Ours (linear)* reduces the Absolute Pose Error (APE) from 1.9845 cm to 1.5238 cm, and *Ours (cubic)* to 1.1867 cm, a 40.2% relative improvement over BAD-4DGau. (w/ EV) and 35.0% over EDyGS. Notably, EDyGS attains competitive photometric scores despite the highest APE among the B-spline and per-pose formulations, indicating that per-pose optimization can fit individual views without recovering a globally consistent trajectory. Our margin widens under shift-invariant evaluation, which supports this interpretation. Per-scene APE and trajectory visualizations across all eight scenes are provided in Appendix A.5 (Table 8, Figure 6).

5.3 Robustness under Increasing Blur

Next, to analyze the performance degradation across increasing blur magnitudes, we evaluate our model under all blur settings: MB, SB, and XB. This experiment quantifies the robustness of our event-aided pipeline, initialized via EDI, to operate effectively at high blur magnitudes.

As shown in Table 4, our method exhibits only a moderate decline in performance as the blur level increases from MB to SB and XB. For example, SI-PSNR drops by 0.67 dB in novel view synthesis between MB and XB settings. These findings confirm that our approach is robust to increasing blur magnitudes. Per-scene results are reported in Appendix A.5 (Table 6, Figure 5).

	Novel View Synthesis		
	SI-PSNR↑	SI-SSIM↑	SI-LPIPS↓
Ours (MB)	21.09	0.5736	0.3792
Ours (SB)	20.75 (−0.34)	0.5604 (−0.0132)	0.4037 (+0.0245)
Ours (XB)	20.42 (−0.67)	0.5503 (−0.0233)	0.4244 (+0.0452)

Table 4: Quantitative evaluation of our method under increasing blur levels: medium (MB), severe (SB), and extreme (XB). Red values show degradation from the MB baseline.

6 Conclusion

We present a dynamic scene reconstruction framework that uses event data to handle extreme motion blur. Our approach introduces a novel continuous B-spline trajectory formulation for the camera pose module. Experiments show consistent gains over previous work on deblurring and novel view synthesis, including cases where conventional pipelines fail. Our comparison of pose formulations further validates the effectiveness of our camera pose module, demonstrating notable improvements in reconstruction quality and reductions in pose error. Our method achieves robust dynamic reconstruction without accurate initial poses, outperforming existing event-based approaches under severe motion blur.

Acknowledgments and Disclosure of Funding

This research is sponsored by the NSF Partnerships for Innovation grant (#2329780). The authors acknowledge Research Computing at Arizona State University for providing computing and storage resources that have contributed to the research results reported in this paper [15].

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A Supplementary Material

A.1 Background: Gaussian Splatting for Dynamic Scene Reconstruction

Gaussian Splatting [16] represents a static scene in the form of a point cloud, where each point is a Gaussian characterized by its center point μ and its covariance matrix Σ . The covariance matrix is split into a scaling vector $\mathbf{s} \in \mathbb{R}^3$, which defines the Gaussian’s size in each direction, and a rotation quaternion $\mathbf{r} \in \mathbb{R}^4$, which defines the orientation. Additionally, each Gaussian has opacity $\alpha \in \mathbb{R}$, and spherical harmonic (SH) coefficients $\mathcal{C} \in \mathbb{R}^K$ (where K represents the number of SH coefficients)

For a dynamic scene, a single set of canonical 3D Gaussians $\mathbb{G}_c$ is learned and maintained throughout the sequence. At each timestamp t , this canonical set $\mathbb{G}_c$ is deformed by a Gaussian deformation field $\Delta\mathbb{G}$, which transforms the mean μ , covariance Σ , opacity α , and color $\mathcal{C}$ of canonical Gaussians over time. The deformation field $\Delta\mathbb{G}$ is obtained using $\mathcal{F}$, modeled using either a neural network [1, 48, 52] or using explicit motion model [12, 24, 26].

$$\Delta\mathbb{G} = \mathcal{F}(\mathbb{G}_c, t) \quad \text{and} \quad \mathbb{G}_t = \mathbb{G}_c + \Delta\mathbb{G}. \quad (7)$$

To render an image at time t , the camera pose is represented by a view matrix $\mathbf{P}(t) = [\mathbf{R} \mid \mathbf{T}]$. The deformed Gaussians $\mathbb{G}_t$ at time t are projected onto the 2D image plane and rasterized to produce the corresponding image $\mathbf{C}(t)$. This rendering process is performed using differentiable splatting [53] $\mathcal{S}$ with $\mathbf{C}(t) = \mathcal{S}(\mathbf{P}(t), \mathbb{G}_t)$.

A.2 Synthetic Blur and Event-data Generation

Given a video sequence of m frames at 30 FPS, we first reserve every 10th frame as a held-out set for novel view synthesis evaluation. The remaining frames are temporally upsampled using RIFE [13], interpolating 7 intermediate frames between each consecutive pair, thereby increasing the frame rate to 240 FPS. To synthesize motion blur, we apply temporal averaging over the upsampled frames. Specifically, we generate 1 extreme blur (XB) sample, 2 severe blur (SB) samples, and 3 medium blur (MB) samples per original frame interval by averaging over 72, 40, and 24 consecutive interpolated frames, respectively. We simulate events from these upsampled sequences using the ESIM [38] simulator. For each level of blur, we reconstruct grayscale sharp intensity images using the Event-based Double Integral (EDI) [34] method, matching the frequency of the original 30 FPS sequence. Camera poses are estimated using COLMAP [40] applied on the EDI-reconstructed images across all blur levels, resulting in three distinct sets of pose estimates per blur level. These are evaluated against the ground truth camera poses and sharp images derived from the original video sequences. Following the procedure above, we use 8 scenes from the DyCheck [10] dataset: *backpack*, *handwavy*, *paper-windmill*, *pillow*, *space-out*, *spin*, *sriracha-tree*, and *teddy* and report results on the full resolution.

To obtain dynamic-region masks, we apply Segment Anything [17] to the blurry images and track the masks across the entire sequence with XMem [4] long-video segmentation.

A.3 Initialization Strategy and Training Hyperparameters

The camera poses and point clouds obtained from COLMAP are subject to varying scales and coordinate origins, making alignment across different reconstructions non-trivial. In particular, the DyCheck dataset provides an initial point cloud from Record3D [41], which differs significantly from the COLMAP-generated reconstructions, leading to failures in standard RANSAC [9] alignment. To address this, we initialize all experiments with a randomly sampled point cloud with 5,000 points. Empirically, we find that this initialization strategy introduces only a minor drop in performance.

We apply Umeyama [19] alignment to all the estimated camera poses using the original camera poses as reference. To ensure stability during pose optimization, we freeze the camera poses for the first 5,000 iterations of the second training stage (out of 10,000 total), preventing divergence from the initially estimated trajectories. Additionally, we adopt hyperparameters consistent with prior work, specifically those used in 4DGaussians [48] and ED3DGS [1].

A.4 Details on Camera Pose Module

Our method uses a cubic B-spline in $\mathbf{SE}(3)$ to model the camera trajectory, with $n + 3$ control points per scene and 7 parameters per control point (pypose Lie tensor), resulting in a parameter tensor of shape $[(n + 3), 7]$. In comparison, the BAD-Gaussians camera pose module assigns 4 control points per frame, leading to $[m, 4, 7]$ parameters for m blurry frames. This fixed allocation causes under-parameterization for longer exposure or complex motion, and over-parameterization in simpler scenes. By decoupling the number of control points from the frame count, our method offers greater flexibility, balancing trajectory smoothness and expressiveness based on scene complexity.

The optimal number of segments for each scene is as follows:

Scene	# Segments
<i>backpack</i>	200
<i>handwavy</i>	200
<i>paper-windmill</i>	200
<i>pillow</i>	240
<i>space-out</i>	280
<i>spin</i>	320
<i>sriracha-tree</i>	320
<i>teddy</i>	200

Table 5: Optimal number of B-spline segments for each scene.

A.5 Per-Scene Results and Analysis



Figure 5: **Qualitative comparison** at medium (MB), severe (SB) and extreme (XB) blur.

Table 6 analyzes performance degradation as blur severity increases from medium to extreme (MB $\rightarrow$ SB $\rightarrow$ XB), demonstrating the robustness of our approach across scenes. Figure 5 shows qualitative comparisons across blur levels and and Table 7 reports per-scene quantitative results on extreme blur (XB). Finally, Table 8 compares our linear and cubic camera pose modules with the MLP-based, BAD-Gaussians, and EDyGS pose formulations under extreme blur, and Figure 6 visualizes the resulting trajectories against BAD-Gaussians.

Scene	Novel View Synthesis		
	SI-PSNR $\uparrow$	SI-SSIM $\uparrow$	SI-LPIPS $\downarrow$
<i>backpack</i> (MB)	19.21	0.3830	0.5961
<i>backpack</i> (SB)	19.12 (−0.09)	0.3813 (−0.002)	0.5911 (−0.005)
<i>backpack</i> (XB)	18.94 (−0.27)	0.3784 (−0.005)	0.5981 (+0.002)
<i>handwavy</i> (MB)	23.75	0.7406	0.2321
<i>handwavy</i> (SB)	22.86 (−0.89)	0.7141 (−0.027)	0.2958 (+0.064)
<i>handwavy</i> (XB)	22.88 (−0.87)	0.7158 (−0.025)	0.2878 (+0.056)
<i>paper-windmill</i> (MB)	17.94	0.3847	0.4726
<i>paper-windmill</i> (SB)	17.83 (−0.11)	0.3738 (−0.011)	0.5171 (+0.045)
<i>paper-windmill</i> (XB)	17.52 (−0.42)	0.3700 (−0.015)	0.5289 (+0.056)
<i>pillow</i> (MB)	18.34	0.5347	0.4270
<i>pillow</i> (SB)	18.13 (−0.20)	0.5267 (−0.008)	0.4292 (+0.002)
<i>pillow</i> (XB)	17.51 (−0.83)	0.5054 (−0.029)	0.4749 (+0.048)
<i>space-out</i> (MB)	24.32	0.7287	0.2626
<i>space-out</i> (SB)	24.14 (−0.18)	0.7244 (−0.004)	0.2759 (+0.013)
<i>space-out</i> (XB)	23.76 (−0.56)	0.7157 (−0.013)	0.2964 (+0.034)
<i>spin</i> (MB)	21.46	0.5327	0.3630
<i>spin</i> (SB)	21.12 (−0.34)	0.5112 (−0.021)	0.4184 (+0.055)
<i>spin</i> (XB)	20.77 (−0.69)	0.4957 (−0.037)	0.4535 (+0.091)
<i>sriracha-tree</i> (MB)	25.94	0.7876	0.2486
<i>sriracha-tree</i> (SB)	25.47 (−0.47)	0.7814 (−0.006)	0.2584 (+0.010)
<i>sriracha-tree</i> (XB)	25.16 (−0.78)	0.7733 (−0.014)	0.2769 (+0.028)
<i>teddy</i> (MB)	17.77	0.4964	0.4316
<i>teddy</i> (SB)	17.35 (−0.42)	0.4706 (−0.026)	0.4436 (+0.012)
<i>teddy</i> (XB)	16.80 (−0.97)	0.4477 (−0.049)	0.4788 (+0.047)

Table 6: Per-scene performance of our method under increasing blur levels — medium (MB), severe (SB), and extreme (XB). Red values show degradation from the MB baseline.

Scene: backpack							Scene: handwavy					
Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	SI-PSNR $\uparrow$	SI-SSIM $\uparrow$	SI-LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	SI-PSNR $\uparrow$	SI-SSIM $\uparrow$	SI-LPIPS $\downarrow$
ED3DGS	18.32	0.3302	0.7050	17.53	0.3125	0.7148	21.86	0.6956	0.4286	20.67	0.6935	0.4235
4DGau.	18.35	0.3344	0.7010	17.71	0.3163	0.7096	22.17	0.6961	0.4267	21.33	<u>0.6938</u>	0.4214
4DGau. + MPRNet	17.43	0.3051	0.7155	17.68	0.3175	0.7032	22.07	0.6922	0.4229	21.03	0.6865	0.4191
DyBluRF	13.77	0.1780	0.9270	13.77	0.1750	0.9270	21.23	0.4040	0.5500	19.49	0.3740	0.5640
BAD-4DGau.	17.35	0.2859	0.6534	16.77	0.2693	0.6653	21.18	0.5898	0.3602	20.13	0.5818	0.3713
4DGau. (w/ EV)	18.58	0.3483	0.6871	<u>17.81</u>	0.3216	0.6985	22.99	0.7170	0.3911	21.51	0.7060	0.3901
BAD-4DGau. (w/ EV)	19.46	<u>0.3886</u>	<u>0.5934</u>	17.69	<u>0.3246</u>	<u>0.6206</u>	<u>23.73</u>	<u>0.7202</u>	<u>0.2983</u>	<u>21.91</u>	0.6907	<u>0.3079</u>
EDyGS	18.91	0.3542	0.7071	14.79	0.2902	0.7578	21.34	0.7023	0.4808	21.10	0.6961	0.4818
Ours	<u>19.42</u>	0.3935	0.5916	18.94	0.3784	0.5981	24.41	0.7402	0.2780	22.88	0.7158	0.2878
Scene: paper-windmill							Scene: pillow					
Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	SI-PSNR $\uparrow$	SI-SSIM $\uparrow$	SI-LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	SI-PSNR $\uparrow$	SI-SSIM $\uparrow$	SI-LPIPS $\downarrow$
ED3DGS	18.34	0.3615	0.6312	18.37	0.3656	0.6267	17.19	0.4978	0.5576	14.19	0.4440	0.6150
4DGau.	18.31	0.3625	0.6440	18.26	0.3643	0.6444	17.40	0.5098	0.5502	14.76	0.4520	0.5978
4DGau. + MPRNet	18.28	0.3619	0.6488	18.22	0.3647	0.6439	17.35	0.5062	0.5415	15.06	0.4547	0.5879
DyBluRF	17.24	0.1300	0.7780	16.59	0.1220	0.7810	15.92	0.2400	0.6550	14.35	0.2140	0.6760
BAD-4DGau.	17.86	0.3425	0.5370	17.68	0.3427	0.5369	15.01	0.3808	0.5510	13.84	0.3630	0.5707
4DGau. (w/ EV)	18.69	0.3761	0.6269	<u>18.39</u>	<u>0.3716</u>	0.6248	17.94	<u>0.5306</u>	0.5474	15.71	0.4782	0.5854
BAD-4DGau. (w/ EV)	19.68	0.3944	0.5284	18.33	0.3674	0.5407	<u>18.15</u>	0.5186	<u>0.4746</u>	<u>15.94</u>	0.4623	<u>0.5126</u>
EDyGS	20.71	0.5733	0.4314	18.88	0.4177	0.4373	17.64	0.5218	0.6139	15.67	<u>0.4880</u>	0.6344
Ours	<u>20.48</u>	<u>0.4281</u>	<u>0.4807</u>	17.52	0.3700	<u>0.5289</u>	18.52	0.5355	0.4588	17.51	0.5054	0.4749
Scene: space-out							Scene: spin					
Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	SI-PSNR $\uparrow$	SI-SSIM $\uparrow$	SI-LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	SI-PSNR $\uparrow$	SI-SSIM $\uparrow$	SI-LPIPS $\downarrow$
ED3DGS	21.40	0.6527	0.4743	20.79	0.6707	0.4595	20.53	0.4731	0.5642	18.75	0.4479	0.5835
4DGau.	21.55	0.6571	0.4706	21.53	0.6767	0.4500	20.60	0.4797	0.5719	18.84	0.4549	0.5919
4DGau. + MPRNet	21.58	0.6556	0.4517	21.56	0.6760	0.4296	20.48	0.4780	0.5691	19.32	0.4576	0.5797
DyBluRF	19.59	0.3650	0.5760	17.93	0.3600	0.5840	16.15	0.1700	0.7800	16.03	0.1660	0.7830
BAD-4DGau.	20.87	0.6197	0.3941	21.22	0.6459	0.3710	19.60	0.4310	0.5277	18.69	0.4188	0.5349
4DGau. (w/ EV)	22.04	0.6687	0.4467	21.74	0.6841	0.4278	21.03	0.4927	0.5483	19.57	0.4693	0.5631
BAD-4DGau. (w/ EV)	22.91	0.6710	<u>0.3447</u>	<u>22.56</u>	<u>0.6889</u>	<u>0.3288</u>	<u>21.65</u>	0.4910	<u>0.4940</u>	<u>20.70</u>	0.4705	<u>0.5070</u>
EDyGS	<u>24.38</u>	0.7165	0.3734	21.30	0.6883	0.3785	20.90	<u>0.4998</u>	0.5833	19.95	<u>0.4732</u>	0.5920
Ours	24.65	<u>0.7056</u>	0.3063	23.76	0.7157	0.2964	22.82	0.5497	0.4310	20.77	0.4957	0.4535
Scene: sriracha-tree							Scene: teddy					
Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	SI-PSNR $\uparrow$	SI-SSIM $\uparrow$	SI-LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	SI-PSNR $\uparrow$	SI-SSIM $\uparrow$	SI-LPIPS $\downarrow$
ED3DGS	23.27	0.7482	0.3655	21.35	0.7307	0.3728	15.85	0.4358	0.5659	13.36	0.3917	0.6210
4DGau.	23.62	0.7574	0.3862	23.23	0.7465	0.3843	16.77	0.4596	0.5678	13.75	0.3990	0.6291
4DGau. + MPRNet	23.67	0.7561	0.3679	23.00	0.7428	0.3680	16.38	0.4513	0.5574	13.64	0.3965	0.6221
DyBluRF	19.95	0.4630	0.5410	19.45	0.4550	0.5460	15.51	0.2240	0.7440	14.76	0.2100	0.7570
BAD-4DGau.	21.82	0.6653	0.3594	21.50	0.6581	0.3628	15.54	0.3586	0.5105	13.20	0.3188	0.5561
4DGau. (w/ EV)	24.31	0.7698	0.3684	23.58	0.7536	0.3667	17.22	<u>0.4701</u>	0.5432	14.28	0.4080	0.6012
BAD-4DGau. (w/ EV)	25.56	0.7726	<u>0.2798</u>	24.10	0.7539	<u>0.2930</u>	16.74	0.4338	<u>0.4736</u>	13.95	0.3776	<u>0.5341</u>
EDyGS	28.19	0.8200	0.3033	26.13	0.7948	0.3088	18.39	0.5169	0.5531	<u>14.79</u>	<u>0.4397</u>	0.5909
Ours	<u>27.07</u>	<u>0.7971</u>	0.2613	<u>25.16</u>	<u>0.7733</u>	0.2769	<u>17.27</u>	0.4543	0.4652	16.80	0.4477	0.4788

Table 7: Per-scene performance comparison across deblurring and novel view synthesis on Extreme Blur (XB).

Scene	Method	Deblurring			Novel View Synthesis			APE (cm)↓
		PSNR↑	SSIM↑	LPIPS↓	SI-PSNR↑	SI-SSIM↑	SI-LPIPS↓	
<i>backpack</i>	MLP-4DGau. (w/ EV)	18.33	0.3302	0.6740	17.93	0.3228	0.6788	2.0171
	BAD-4DGau. (w/ EV)	19.46	0.3886	0.5934	17.69	0.3246	0.6206	2.1904
	EDyGS	18.91	0.3542	0.7071	14.79	0.2902	0.7578	2.5814
	Ours (linear)	18.37	0.3553	0.6765	12.40	0.2528	0.7531	1.9297
	Ours (cubic)	19.42	0.3935	0.5916	18.94	0.3784	0.5981	1.4705
<i>handwavy</i>	MLP-4DGau. (w/ EV)	21.67	0.6455	0.3798	21.09	0.6524	0.3764	1.2255
	BAD-4DGau. (w/ EV)	23.73	0.7202	0.2983	21.91	0.6907	0.3079	1.0646
	EDyGS	21.34	0.7023	0.4808	21.10	0.6961	0.4818	1.4346
	Ours (linear)	24.45	0.7404	0.2646	23.87	0.7413	0.2699	0.4057
	Ours (cubic)	24.41	0.7402	0.2780	22.88	0.7158	0.2878	0.4274
<i>paper-windmill</i>	MLP-4DGau. (w/ EV)	18.04	0.3474	0.5914	18.14	0.3550	0.5851	1.1202
	BAD-4DGau. (w/ EV)	19.68	0.3944	0.5284	18.33	0.3674	0.5407	1.1462
	EDyGS	20.71	0.5733	0.4314	18.88	0.4177	0.4373	1.6557
	Ours (linear)	20.55	0.4325	0.4658	20.21	0.4424	0.4677	0.2833
	Ours (cubic)	20.48	0.4281	0.4807	17.52	0.3700	0.5289	0.328
<i>pillow</i>	MLP-4DGau. (w/ EV)	16.27	0.4348	0.5359	15.05	0.4152	0.5571	2.1436
	BAD-4DGau. (w/ EV)	18.15	0.5186	0.4746	15.94	0.4623	0.5126	2.7079
	EDyGS	17.64	0.5218	0.6139	15.67	0.4880	0.6344	2.3740
	Ours (linear)	18.54	0.5480	0.4391	18.26	0.5306	0.4499	1.7422
	Ours (cubic)	18.52	0.5355	0.4588	17.51	0.5054	0.4749	1.7771
<i>space-out</i>	MLP-4DGau. (w/ EV)	20.82	0.6346	0.4170	21.80	0.6716	0.3871	2.6245
	BAD-4DGau. (w/ EV)	22.91	0.6710	0.3447	22.56	0.6889	0.3288	2.7376
	EDyGS	24.38	0.7165	0.3734	21.30	0.6883	0.3785	2.1002
	Ours (linear)	24.50	0.7022	0.2978	25.47	0.7426	0.2734	1.9934
	Ours (cubic)	24.65	0.7056	0.3063	23.76	0.7157	0.2964	2.0324
<i>spin</i>	MLP-4DGau. (w/ EV)	19.73	0.4458	0.5383	19.44	0.4437	0.5431	1.8787
	BAD-4DGau. (w/ EV)	21.65	0.4910	0.4940	20.70	0.4705	0.5070	2.0566
	EDyGS	20.90	0.4998	0.5833	19.95	0.4732	0.5920	1.3033
	Ours (linear)	23.12	0.5691	0.4144	21.77	0.5423	0.4293	1.2489
	Ours (cubic)	22.82	0.5497	0.4310	20.77	0.4957	0.4535	1.2603
<i>sriracha-tree</i>	MLP-4DGau. (w/ EV)	22.87	0.7245	0.3346	23.40	0.7297	0.3336	2.256
	BAD-4DGau. (w/ EV)	25.56	0.7726	0.2798	24.10	0.7539	0.2930	2.203
	EDyGS	28.19	0.8200	0.3033	26.13	0.7948	0.3088	1.4707
	Ours (linear)	24.41	0.7558	0.3169	13.83	0.6372	0.4815	3.0319
	Ours (cubic)	27.07	0.7971	0.2613	25.16	0.7733	0.2769	0.6211
<i>teddy</i>	MLP-4DGau. (w/ EV)	15.73	0.3776	0.5126	13.88	0.3495	0.5473	1.8843
	BAD-4DGau. (w/ EV)	16.73	0.4338	0.4736	13.95	0.3776	0.5341	1.7701
	EDyGS	18.39	0.5169	0.5531	14.79	0.4397	0.5909	1.6991
	Ours (linear)	16.98	0.4431	0.4711	17.07	0.4510	0.4750	1.5556
	Ours (cubic)	17.27	0.4543	0.4652	16.80	0.4477	0.4788	1.5769

Table 8: Per-scene comparison of our camera pose module and that of BAD Gaussians[56] on view synthesis and camera trajectory estimate in extreme blur (XB) setting.

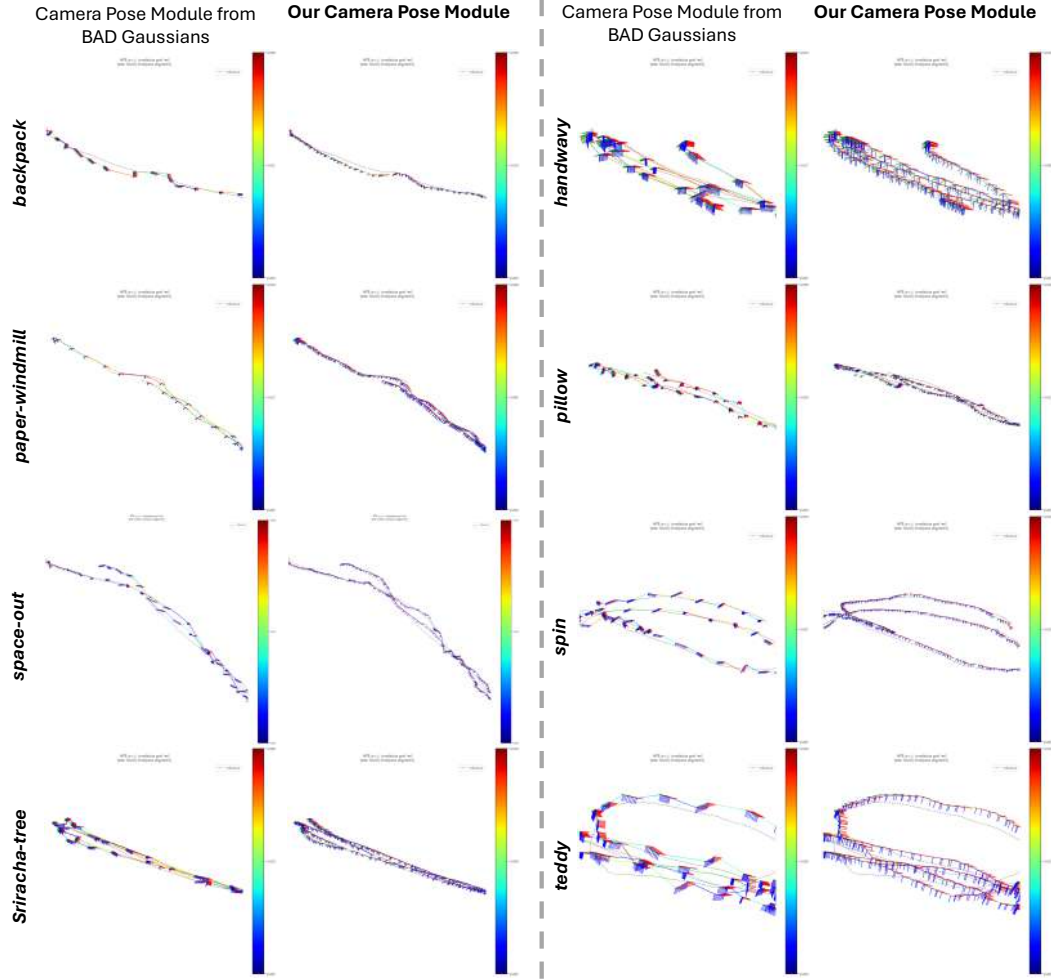


Figure 6: **Estimated camera trajectories.** Comparison between the BAD Gaussians pose module [56] and our pose module. The BAD Gaussians module yields discontinuous, non-smooth trajectories while our method produces smoother camera trajectory across all scenes.